

ECO 310: Precept 4

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Outline

- 1 Question 1: Constant Absolute Risk Aversion
- 2 Question 2: Risk-Lover's Willingness to Pay
- 3 Question 3: Betting on the Chiefs (Eagles Fan)

Q1: Setup

Problem Statement:

Matt has constant absolute risk aversion with

$$u(w) = 1 - e^{-\alpha w}, \quad \alpha > 0.$$

He is offered a gamble:

$$\begin{cases} \text{Lose } L & \text{w.p. } \frac{1}{2}, \\ \text{Win } H & \text{w.p. } \frac{1}{2}. \end{cases}$$

Task: Find a threshold L^* such that for $L > L^*$, he is unwilling to take the gamble even if $H \rightarrow \infty$.

Q1: Key Concepts

- **Constant Absolute Risk Aversion (CARA):** With an exponential utility function, risk attitudes do not change with wealth.
- **Expected Utility:** When deciding on a gamble, Matt compares

$$EU = \frac{1}{2}u(w_0 - L) + \frac{1}{2}u(w_0 + H)$$

with the utility of not taking the gamble, $u(w_0)$.

- **Threshold:** The idea is to find L^* so that even if H is arbitrarily high, the gamble is rejected.

Q1: Step 1 – Compute Expected Utility if Accepting the Gamble

Assume initial wealth w_0 . Then if Matt accepts the gamble,

$$EU = \frac{1}{2} \left[1 - e^{-\alpha(w_0 - L)} \right] + \frac{1}{2} \left[1 - e^{-\alpha(w_0 + H)} \right].$$

Simplify:

$$EU = 1 - \frac{1}{2} e^{-\alpha(w_0 - L)} - \frac{1}{2} e^{-\alpha(w_0 + H)}.$$

Q1: Step 2 – Set Up the Rejection Condition

If he does not take the gamble, his utility is

$$u(w_0) = 1 - e^{-\alpha w_0}.$$

He rejects the gamble if

$$1 - e^{-\alpha w_0} \geq 1 - \frac{1}{2}e^{-\alpha(w_0-L)} - \frac{1}{2}e^{-\alpha(w_0+H)}.$$

Cancelling 1 from both sides, we have

$$e^{-\alpha w_0} \leq \frac{1}{2}e^{-\alpha(w_0-L)} + \frac{1}{2}e^{-\alpha(w_0+H)}.$$

Q1: Step 3 – Simplify and Take $H \rightarrow \infty$

Factor out $e^{-\alpha w_0}$:

$$e^{-\alpha w_0} \leq \frac{1}{2} e^{-\alpha w_0} e^{\alpha L} + \frac{1}{2} e^{-\alpha w_0} e^{-\alpha H}.$$

Cancel $e^{-\alpha w_0}$ (which is positive):

$$1 \leq \frac{1}{2} e^{\alpha L} + \frac{1}{2} e^{-\alpha H}.$$

As $H \rightarrow \infty$, $e^{-\alpha H} \rightarrow 0$, so

$$1 \leq \frac{1}{2} e^{\alpha L}.$$

Q1: Step 4 – Find the Threshold L^*

Multiply both sides by 2:

$$2 \leq e^{\alpha L}.$$

Taking logarithms:

$$\alpha L \geq \ln 2 \quad \implies \quad L \geq \frac{\ln 2}{\alpha}.$$

Threshold: $L^* = \frac{\ln 2}{\alpha}.$

Q2: Setup

Problem Statement:

A risk-lover with \$400 expects to gain \$200 with chance $\frac{1}{6}$ and lose \$100 with chance $\frac{5}{6}$.

Task: Show that this individual will pay at most \$50 to avoid this risk.

Q2: Key Concepts

- An individual is **risk-loving** if they prefer a risky lottery to a sure outcome when both have the same expected value.
- In utility terms, for a lottery X , a risk-lover satisfies

$$E[u(X)] > u(E[X]).$$

- Their preferences are typically represented by a *convex* utility function.
- Such individuals derive additional satisfaction from the uncertainty itself.

Q2: Step 1 – Compute Expected Wealth Change

Calculate the expected change:

$$\Delta w = \frac{1}{6}(+200) + \frac{5}{6}(-100) = \frac{200}{6} - \frac{500}{6} = -\frac{300}{6} = -50.$$

Thus, the expected loss is \$50.

Q2: Step 2 – Relate to Willingness to Pay

- A risk-neutral person would pay exactly the expected loss, \$50, to avoid the risk.
- Since this individual is risk-loving (and hence derives extra utility from taking risks), they would be willing to pay *at most* \$50.

Q2: Alternative Mathematical Explanation

Let x be the amount the person is willing to pay to avoid the gamble. Then, avoiding the gamble gives wealth $400 - x$, so

$$u(400 - x) \geq \frac{1}{6}u(600) + \frac{5}{6}u(300).$$

Since for a risk-lover expected utility exceeds the utility of the expected wealth,

$$u(400 - x) \geq u\left(\frac{1}{6} \cdot 600 + \frac{5}{6} \cdot 300\right) = u(350).$$

Therefore, to satisfy $400 - x \geq 350$ we must have

$$x \leq 50.$$

Q3: Setup

Problem Statement:

Imagine you are an Eagles fan considering betting on the Superbowl. The moneyline for betting on the Chiefs is -120 , meaning you must bet \$120 to win \$100. Equivalently, the return rate is

$$r = \frac{100}{120} = \frac{5}{6}.$$

You expect the Chiefs to win with probability $p > \frac{1}{2}$.

Assumptions:

- If the Chiefs win, it is like incurring a loss L (reflecting the disappointment of an Eagles loss), but you gain $\frac{5}{6}x$ from your bet x .
- If the Chiefs lose (i.e. Eagles win), you lose your bet x .
- You care only about wealth with $u(w) = \ln w$.

Task: Write the equation describing your optimal bet x and explain why if the odds are unfavorable (i.e. $\frac{p}{1-p} < \frac{6}{5}$) then you'll bet a positive amount on the Chiefs—but a small enough amount that you still prefer the Eagles win.

Q3: Step 1 – Define Wealth Outcomes and Expected Utility

Let x be the bet on the Chiefs.

- **If Chiefs win:** Final wealth is

$$w_0 - L + \frac{5}{6}x.$$

- **If Chiefs lose:** Final wealth is

$$w_0 - x.$$

Thus, the expected utility is

$$EU(x) = p \ln\left(w_0 - L + \frac{5}{6}x\right) + (1 - p) \ln(w_0 - x).$$

Q3: Step 2 – Derive the First Order Condition (FOC)

Differentiate $EU(x)$ with respect to x :

$$EU'(x) = p \frac{\frac{5}{6}}{w_0 - L + \frac{5}{6}x} - (1 - p) \frac{1}{w_0 - x} = 0.$$

Rearranging gives:

$$\frac{5}{6} \frac{p}{w_0 - L + \frac{5}{6}x} = \frac{1 - p}{w_0 - x}.$$

Q3: Step 3 – Rearrangement and Interpretation

Multiply both sides by $(w_0 - L + \frac{5}{6}x)(w_0 - x)$:

$$\frac{5}{6}p(w_0 - x) = (1 - p)(w_0 - L + \frac{5}{6}x).$$

Divide both sides by $(1 - p)(w_0 - x)$ (assuming $w_0 - x > 0$):

$$\frac{5}{6} \frac{p}{1 - p} = \frac{w_0 - L + \frac{5}{6}x}{w_0 - x}.$$

Notice that if $\frac{p}{1-p} < \frac{6}{5}$, then the left-hand side is less than 1. Hence,

$$w_0 - L + \frac{5}{6}x < w_0 - x,$$

meaning the outcome when the Chiefs win is strictly less than when the Eagles win. In other words, even if you bet a positive amount x (acting as partial insurance against an undesirable outcome), you still prefer the Eagles win.

Q3: Final Remarks on the Optimal Bet

- The FOC

$$\frac{5}{6} \frac{p}{1-p} = \frac{w_0 - L + \frac{5}{6}x}{w_0 - x}$$

describes the optimal x .

- When the odds are unfavorable ($\frac{p}{1-p} < \frac{6}{5}$), the optimal bet x is small. This reflects the intuition that, although betting on the Chiefs provides some insurance (a partial hedge against the loss L), you still prefer the outcome where the Eagles win.

Thursday Mar 6, Online
See you after spring break!